Horizontal and vertical dilations

1 Consider the functions $f(x) = \cos x$ and $g(x) = \cos 4x$.

- **a** State the period of $f(x)$.
- **b** Complete the table of values for $g(x)$.

x	0	22.5	45	67.5	90	112.5	135	157.5	180
$g(x)$									

- **c** State the period of $g(x)$.
- **d** Describe the transformation required to obtain the graph of $g(x)$ from $f(x)$.
- **e** Sketch the graph of $g(x)$ for $0 \leq x \leq 180$.

2 Consider the function $y = 2\cos 3x$.

- **a** State the amplitude of the function.
- **b** Find the period of the function.
- **c** Sketch a graph of the function for $-180 \leq x \leq 180$.

3 For each of the following functions:

- **i** State the amplitude.
- **ii** Find the period.
- **iii** Sketch a graph of the function for $0 \leq x \leq 360$.

a $y = \cos 3x$ **b** $y = \sin\left(\frac{x}{2}\right)$ **c** $y = \sin\left(\frac{3x}{4}\right)$ **d** $y = \cos\left(\frac{x}{3}\right)$

e $y = -\cos 4x$ **f** $y = 3\sin(0.25x)$ **g** $y = 2\sin 3x$ **h** $y = -5\sin 3x$

4 Consider the function $y = \sin\left(\frac{2x}{3}\right)$.

- **a** State the amplitude of the function.
- **b** Find the period of the function in degrees.
- **c** Sketch a graph of the function for $0^\circ \leq x \leq 720^\circ$.

5 A table of values for the the first period of the graph $y = \sin x$ for $x \geq 0$ is given in the first table on the right:

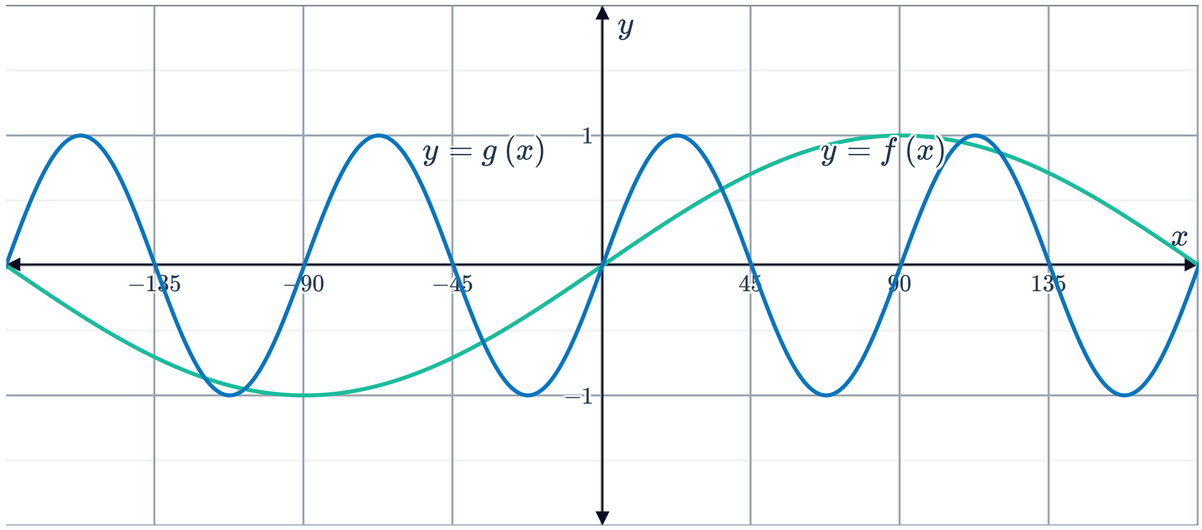
x	0	90	180	270	360
$\sin x$	0	1	0	–1	0

a Complete the second table given with equivalent values for x in the the first period of the graph $y = \sin\left(\frac{x}{4}\right)$ for $x \geq 0$.

x					
$\sin\left(\frac{x}{4}\right)$	0	1	0	–1	0

b Hence, state the period of $y = \sin\left(\frac{x}{4}\right)$.

6 The functions $f(x)$ and $g(x) = f(kx)$ have been graphed on the same set of axes below.



a Describe the transformation required to obtain the graph of $g(x)$ from the graph of $f(x)$.

b Find the value of k.

Combined translations and dilations

7 Consider the function $y = \cos 3x + 2$.

a Find the period of the function.

b State the amplitude of the function.

c Find the maximum value of the function.

d Find the minimum value of the function.

e Sketch a graph of the function for $0 \leq x \leq 360$.

8 For each of the following functions:

- **i** State the domain of the function.
- **ii** State the range of the function.
- **iii** Sketch a graph of the function for $-180 \leq x \leq 180$.

a $y = \sin 2x - 2$

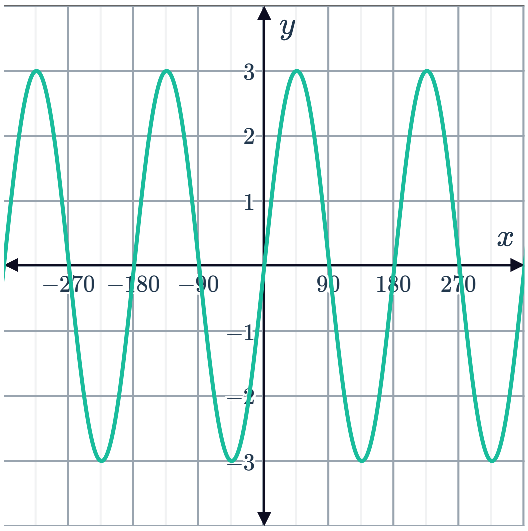
b $y = -5\sin 2x$

c $y = \sin\left(\frac{x}{3}\right) + 5$

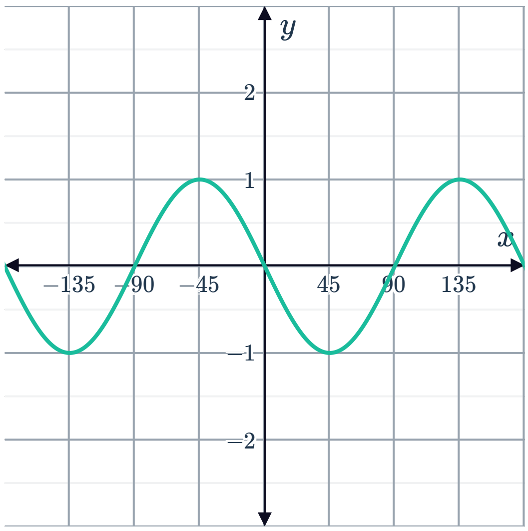
d $y = \cos\left(\frac{x}{2}\right) - 3$

9 For each of the following graphs, find the equation of the function given that it is of the form $y = a\sin bx$ or $y = a\cos bx$, where b is positive:

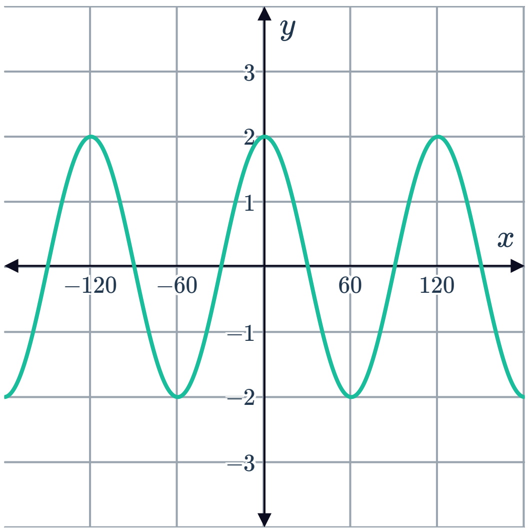
a



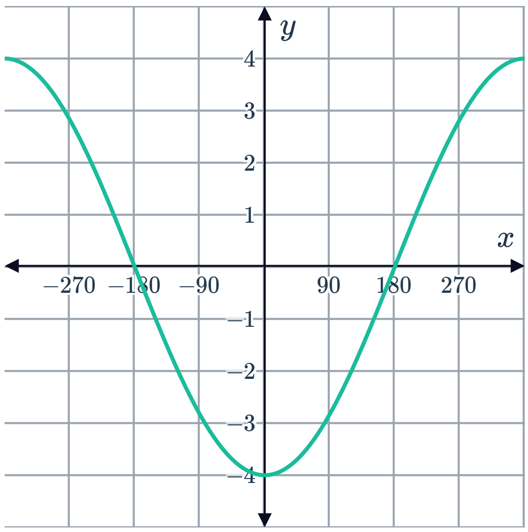
b



c



d



10 State whether the following functions represent a change in the period from the function $y = \sin x$:

a $y = \sin(5x)$

b $y = 5\sin x$

c $y = \sin\left(\frac{x}{5}\right)$

d $y = \sin x + 5$